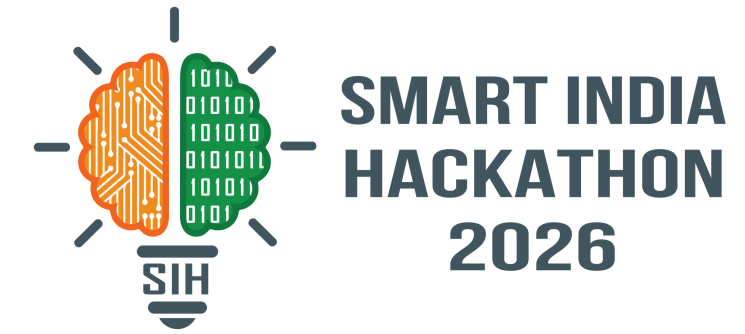


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26166**
- **Problem Statement Title- Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)**
- **Theme- Space Technology**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Horizon**



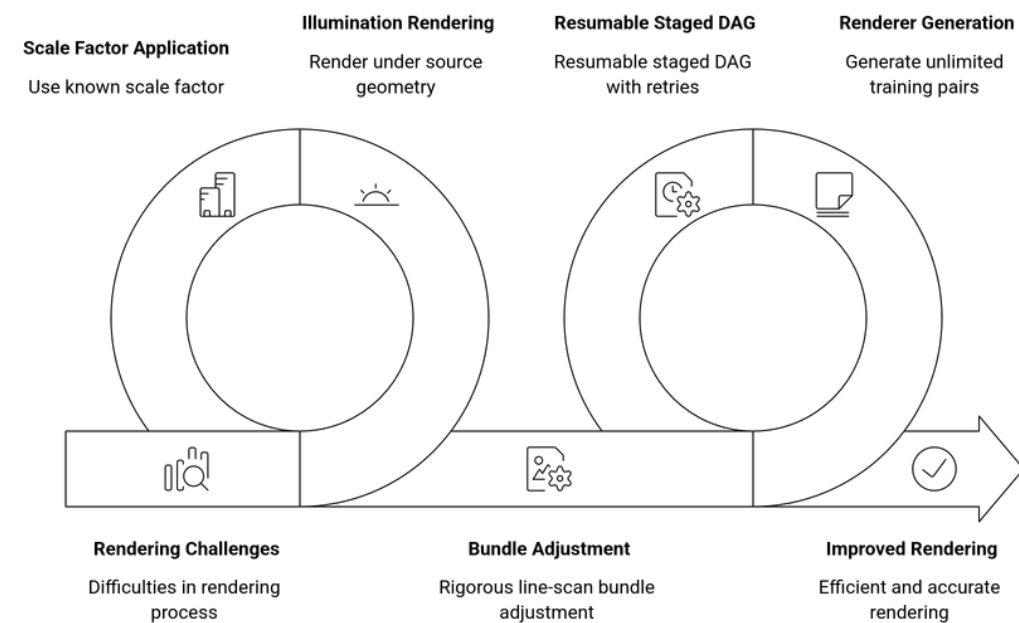
THE PROBLEM: CHANDRAYAAN-2 AND LRO DO NOT OVERLAY

- Telemetry geometry is off by 50–300 m — hundreds of pixels at OHRC's 0.25 m GSD
- Illumination: no atmosphere, so a crater at 12° sun and at 65° sun share almost no gradient structure
- Modality + scale: OHRC 0.25 m, TMC-2 5 m, IIRS ~80 m, NAC 0.5–2 m — GSD ratio up to 1:320
- Geometry: pushbroom line scanners over relief, plus jitter — no affine or projective transform exists
- Today: hours of manual tie-point picking, tens of clustered points, no uncertainty, not reproducible

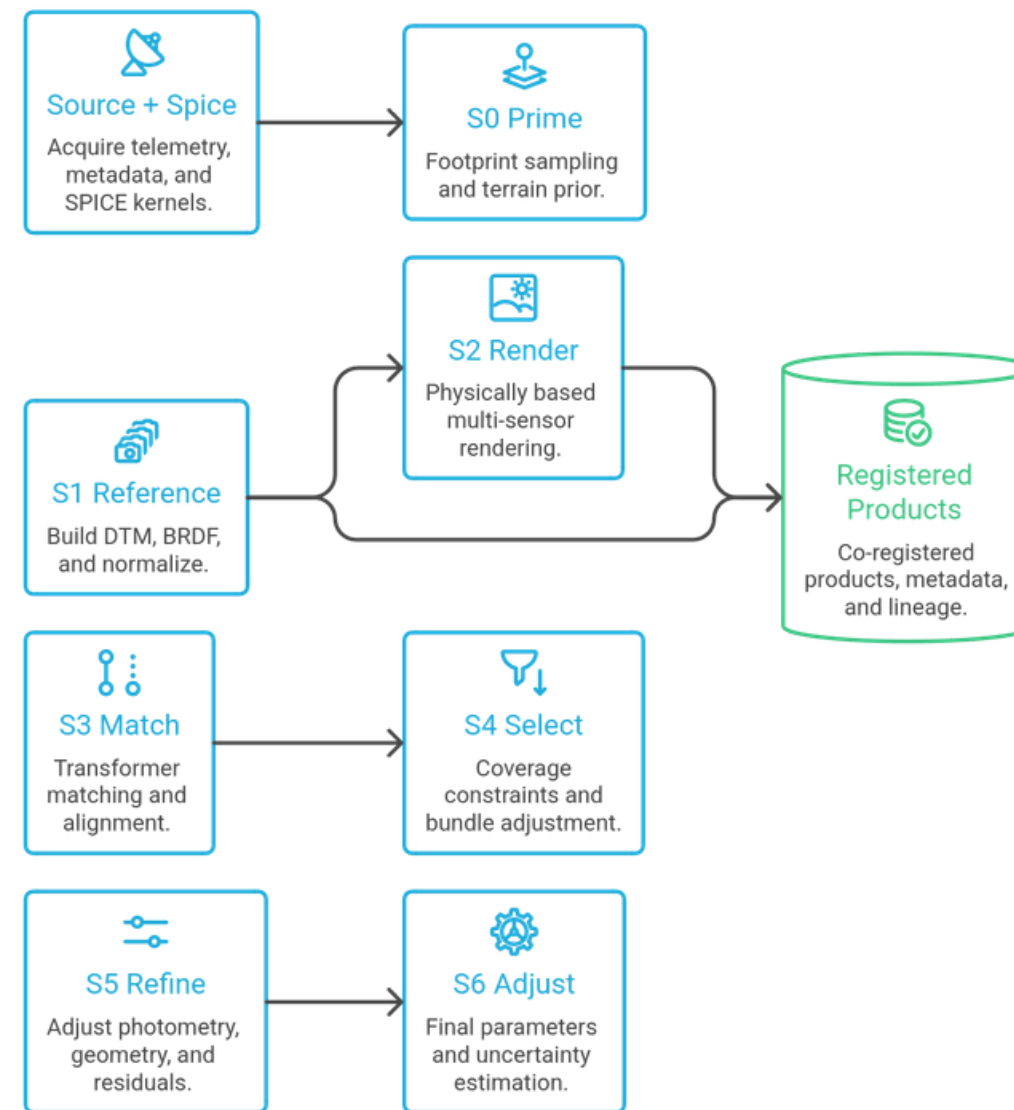
THE IDEA: MODEL THE LIGHT INSTEAD OF FIGHTING IT

- Core inversion: don't seek illumination-invariant descriptors — model the light and cancel it
- S0 Prime: SPICE + line-scan CSM models give footprint, known scale ratio and a dense prior displacement field
- S2 Render: LRO DTM + Hapke BRDF + ray-marched shadows recreate the reference under the source's lighting
- S3 Match: transformer matcher (LoFTR / RoMa lineage) over a five-channel representation stack
- S4–S6: coverage-constrained selection → sub-pixel refinement → Ceres bundle adjustment with jitter
- Scale is never searched: it is read from metadata and applied with sensor-specific PSF matching

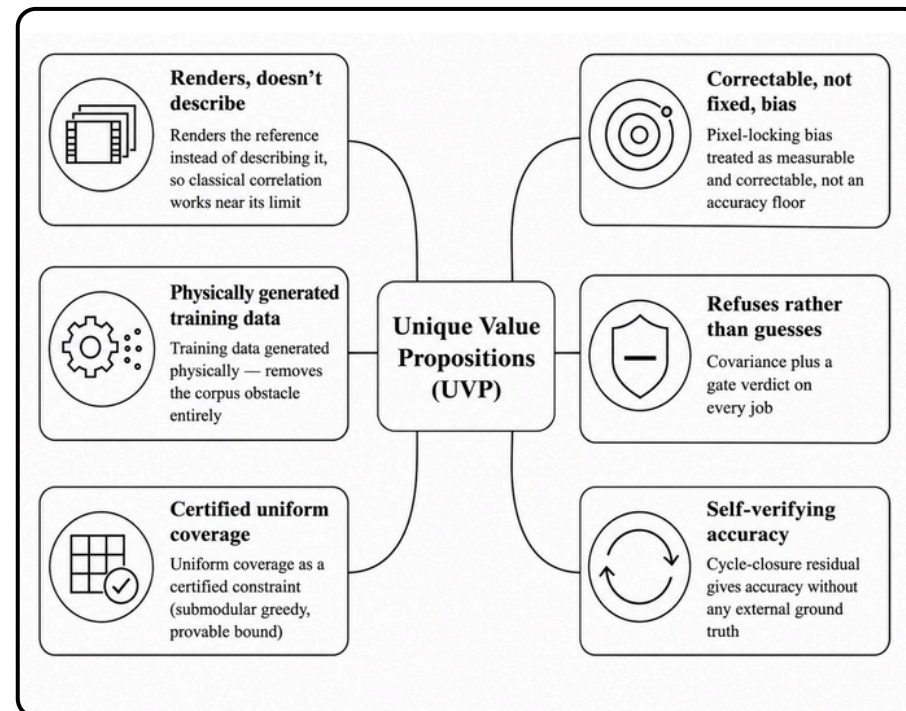
HOW IT ADDRESSES THE PROBLEM



WORKFLOW



- **Pipeline purpose:** A remote-sensing workflow that turns spacecraft telemetry/SPICE data into co-registered terrain and imagery products.
- **Core stages:** Reference building → physically-based rendering → matching/alignment → refinement → final parameter/uncertainty estimation.
- **End goal:** All paths converge into "Registered Products" — traceable outputs with metadata and lineage.



INNOVATION

- Renders the reference instead of describing it, so classical correlation works near its limit
- Training data generated physically — removes the corpus obstacle entirely
- Uniform coverage as a **certified constraint** (submodular greedy, provable bound)
- Pixel-locking bias treated as measurable and correctable, not an accuracy floor
- **Refuses rather than guesses:** covariance plus a gate verdict on every job
- Cycle-closure residual gives accuracy **without any external ground truth**

DATA & UI

PostgreSQL 16 + PostGIS 3.4 (lunar SRID), MinIO COG/Zarr, React 18, OpenLayers 9, titiler

API & SERVICE

Python 3.11, FastAPI, Pydantic v2, Uvicorn behind Nginx; modular monolith

WORKERS & QUEUE

Celery 5.4 in CPU / GPU / IO classes; RabbitMQ 3.13; Redis 7

RENDER & SOLVE

CUDA renderer with OptiX 8 BVH shadows; Ceres 2.2 via pyceres, analytic Jacobians

GEODESY & PHOTOGRAMMETRY

USGS ISIS 8, Ames Stereo Pipeline, usgscsm / csmapi, SpiceyPy, GDAL 3.8

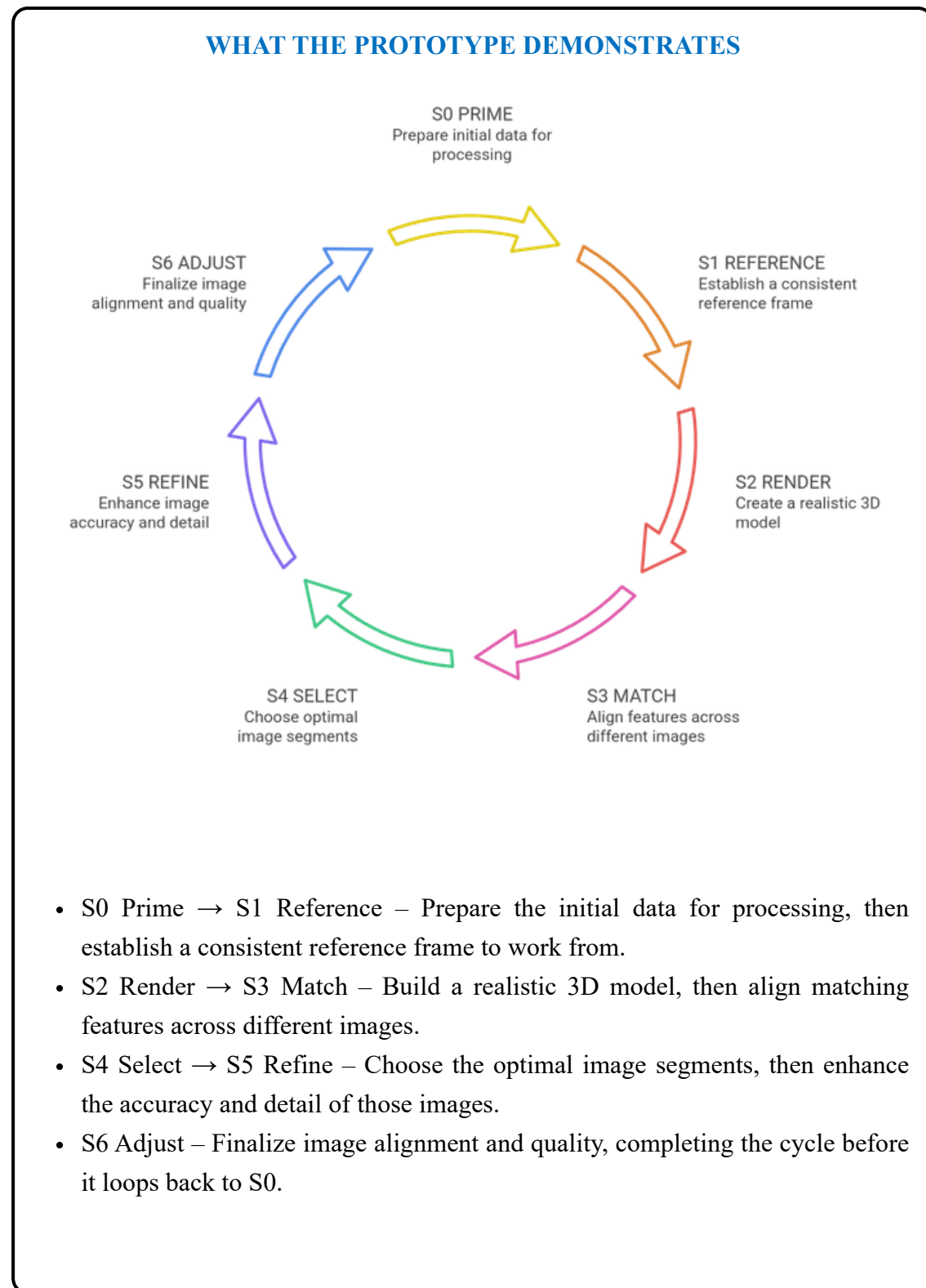
ML & TRAINING

PyTorch 2.4, CUDA 12.4, ONNX Runtime; MLflow and DVC for corpus versioning

Tech-Stack

- **Data & UI** – PostgreSQL 16 + PostGIS 3.4 (lunar SRID) for spatial data, MinIO for COG/Zarr storage, and a React 18 + OpenLayers 9 frontend.
- **API & Service** – Python 3.11 with FastAPI and Pydantic v2, run via Uvicorn behind Nginx as a modular monolith.
- **Workers & Queue** – Celery 5.4 handles CPU/GPU/IO task classes, backed by RabbitMQ 3.13 and Redis 7.
- **Geodesy & Photogrammetry / Render & Solve** – USGS ISIS 8, Ames Stereo Pipeline, usgscsm/csmapi, SpiceyPy, and GDAL 3.8 for geospatial processing; rendering uses a CUDA-based renderer with OptiX 8 BVH shadows and Ceres 2.2 (via pyceres) for analytic Jacobian solves.
- **ML & Training** – PyTorch 2.4 with CUDA 12.4 and ONNX Runtime for inference, plus MLflow and DVC for experiment tracking and corpus/data versioning.

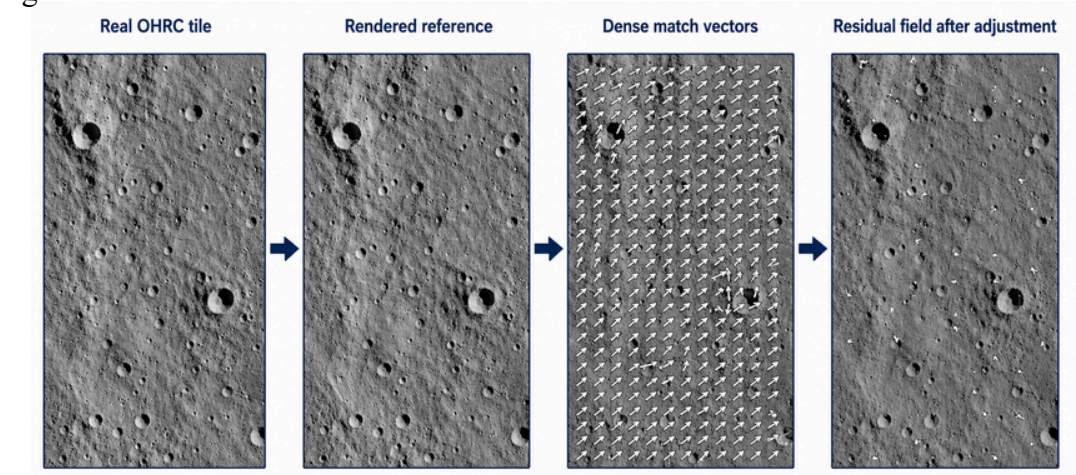
WHAT THE PROTOTYPE DEMONSTRATES



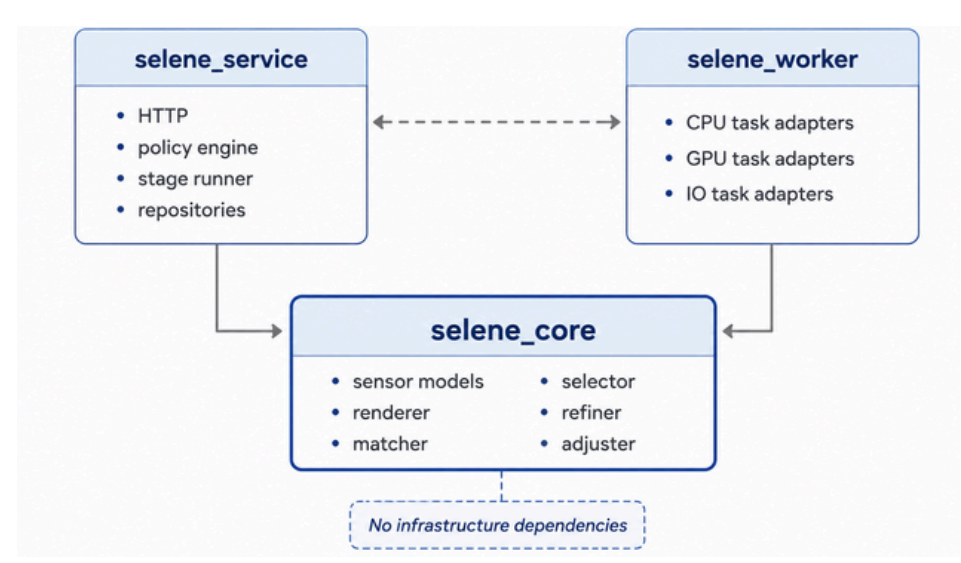
- S0 Prime → S1 Reference – Prepare the initial data for processing, then establish a consistent reference frame to work from.
- S2 Render → S3 Match – Build a realistic 3D model, then align matching features across different images.
- S4 Select → S5 Refine – Choose the optimal image segments, then enhance the accuracy and detail of those images.
- S6 Adjust – Finalize image alignment and quality, completing the cycle before it loops back to S0.

WHAT THE PROTOTYPE DEMONSTRATES

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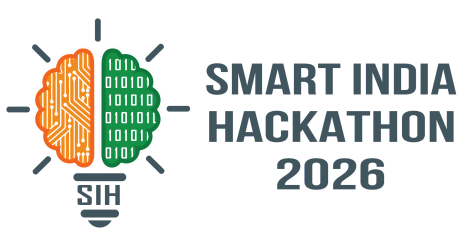


COMPONENT ARCHITECTURE DIAGRAM





FEASIBILITY AND VIABILITY



TECHNICAL FEASIBILITY

- Proven components: ISIS 8, ASP, CSM, SPICE, Ceres, Hapke photometry
- Reference data public and mirrorable: NAC, DTM, SLDEM2015, LOLA
- Telemetry prior (~100s of m) is strong enough to bound the search
- No corpus needed — the renderer supplies supervision
- Core is a dependency-free library, testable on one laptop

OPERATIONAL FEASIBILITY

- On-prem Kubernetes with GPU nodes; fully air-gap capable by design
- 40 OHRC-equivalent scenes per 24 h on 8 GPU workers
- Resumable staged DAG: retries, cancellation, content-addressed cache
- $\geq 97\%$ of jobs reach a terminal state with no operator action
- Roles for analyst, operator, QA and calibration via Keycloak OIDC

ECONOMIC FEASIBILITY

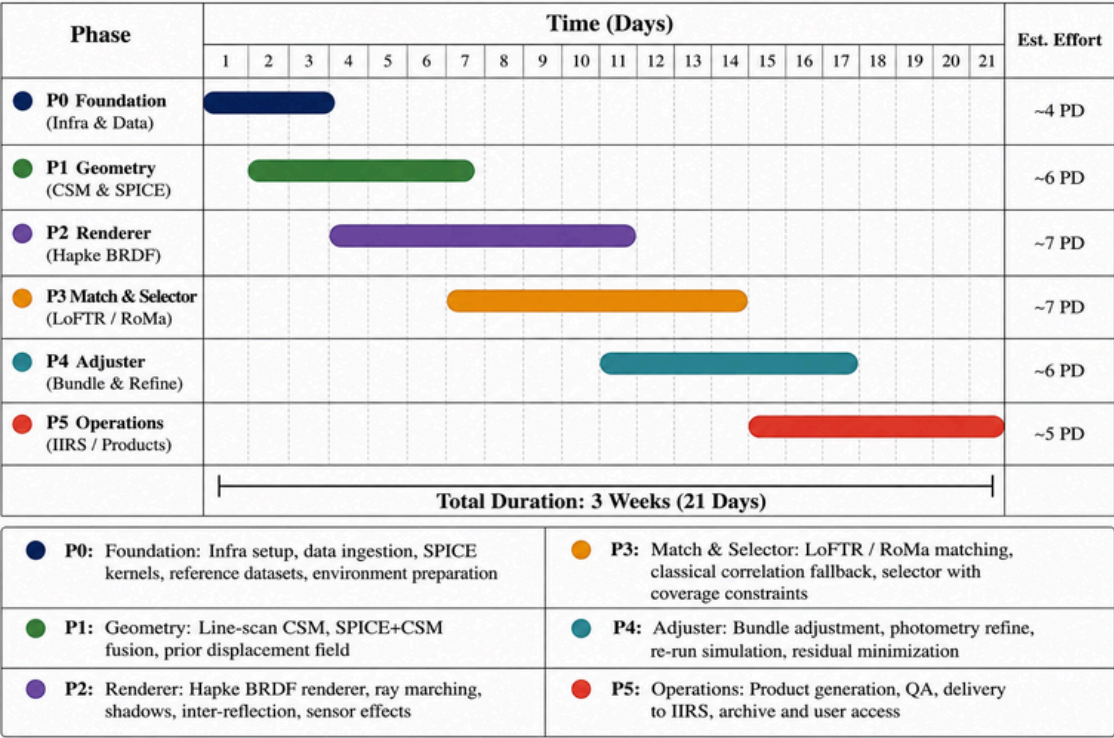
- Open-source throughout — no licence cost anywhere in the stack
- Dev profile: one workstation, one GPU (≥ 40 GB VRAM)
- Renders are cached and shared, so repeat work is near-free
- Replaces hours of analyst time per scene with minutes unattended
- Air-gapped bundle means no recurring cloud spend

VIABILITY & SCALE-UP

- **Sustainability:** open-source, on-premises, single institutional tenant — runs inside ISRO’s own data centre with no external dependency
- **Scale path:** OHRC and TMC-2 → IIRS chained via TMC-2 → multi-payload graph with LRO as datum → Chandrayaan-1 and DFSAR later
- **Durability:** dependency-free scientific core, analytic Jacobians verified in CI, ADRs for every non-obvious choice, STAC catalogue export

KEY RISKS AND MITIGATION

Risk	Impact	Mitigation strategy
R-01 Render not faithful enough for correlation	Med / Critical	Phase-2 go/no-go gate; three channels, so no single one carries the load
R-02 Sim-to-real gap in the learned matcher	Med / High	Domain randomisation, self-supervised fine-tuning, classical fallback live
R-03 Reference DTM quality is the binding limit	High / High	SFS-assist mode; relief diagnostic; DTM uncertainty into covariance
R-04 Kernels weaker than assumed, prior too loose	Med / High	Phase 1 measures prior quality first; coarse pre-alignment fallback
R-06 Runtime exceeds targets at OHRC scale	Med / Med	Shared render cache, NVMe tiling, per-stage perf gate in CI



- **Structure** — 21 days, six overlapping phases (P0–P5). SPICE, Hapke BRDF, and IIRS point to a lunar/planetary remote-sensing pipeline on a tight sprint schedule.
- **P0 + P1 overlap** — P0 (Foundation, days 1–3, ~4 PD) and P1 (Geometry, days 2–7, ~6 PD) overlap almost immediately — geometry only needs core infra, not full foundation completion.
- **Renderer is the biggest block** — P2 (Hapke BRDF rendering, days 4–11, ~7 PD) is the longest single phase and, given its overlap with P1 and P3, the likely critical-path bottleneck.
- **Match + Adjust form the solve loop** — P3 (Matching, days 7–14, 7 PD) and P4 (Adjuster, days 11–17, 6 PD) overlap, suggesting iterative handoff rather than a strict wait between them.
- **Operations tails off with overlap** — P5 (days 14/15–21, ~5 PD) overlaps P4, implying staged QA/delivery on finished batches while other data is still being refined.
- **Effort implies parallel staffing** — ~35 PD of total work packed into 21 days only works with multiple people/workstreams running at once, especially days 4–14 when four phases overlap.

TARGET AUDIENCE

- **Planetary scientists** registering scenes, exporting tie points
- **Production operators** running bulk registration campaigns
- **QA reviewers** accepting or rejecting against numeric gates
- **Calibration engineers** tuning sensor and photometric parameters
- **Downstream pipelines:** change detection, DTM, site selection

SCIENTIFIC & NATIONAL IMPACT

- Turns the Chandrayaan-2 archive into one co-registered dataset
- Zero silent wrong answers: covariance + verdict on every job
- Reproducible to 1e-6 px, so analysts get identical results
- Strengthens India's lunar geodetic capability
- Synthetic corpus published as a community benchmark

ECONOMIC BENEFITS

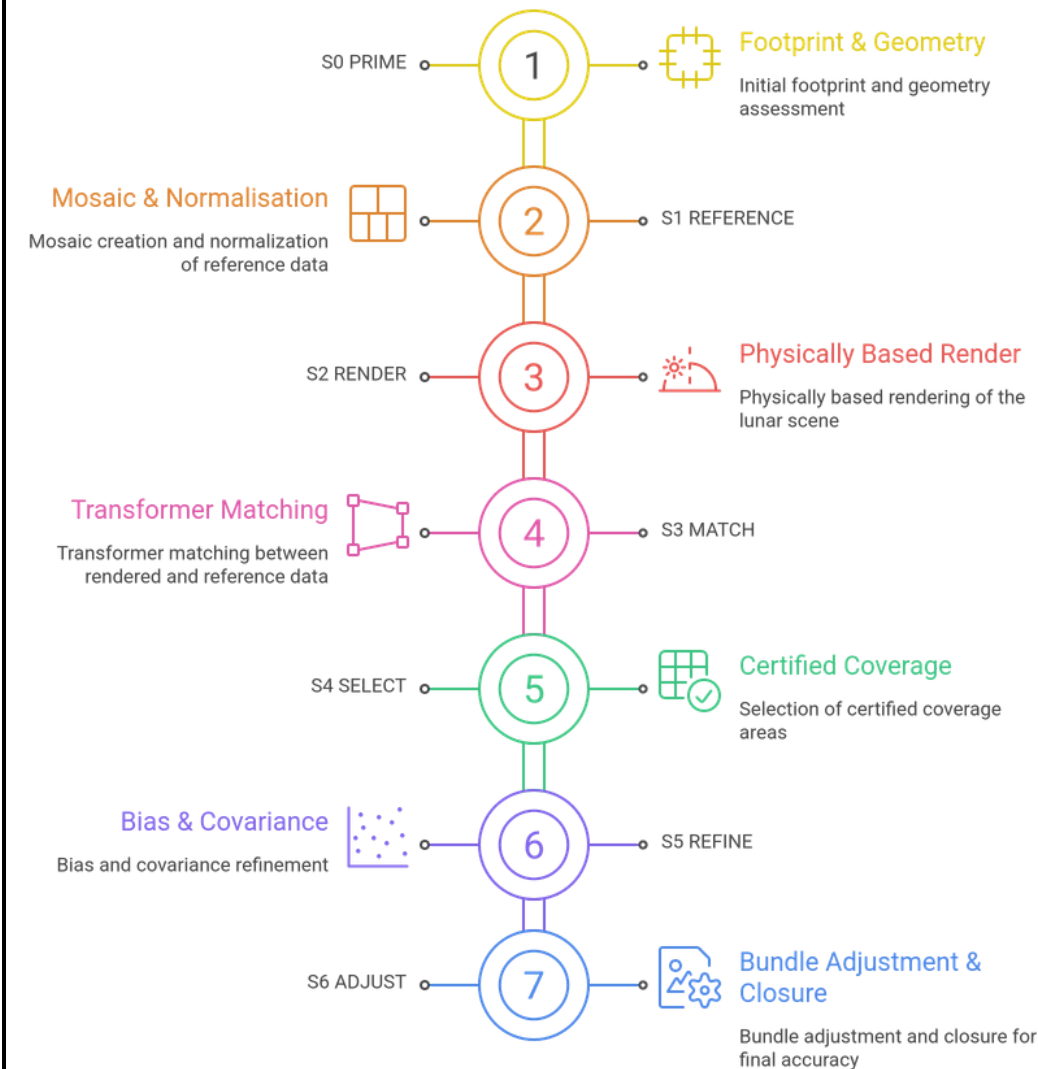
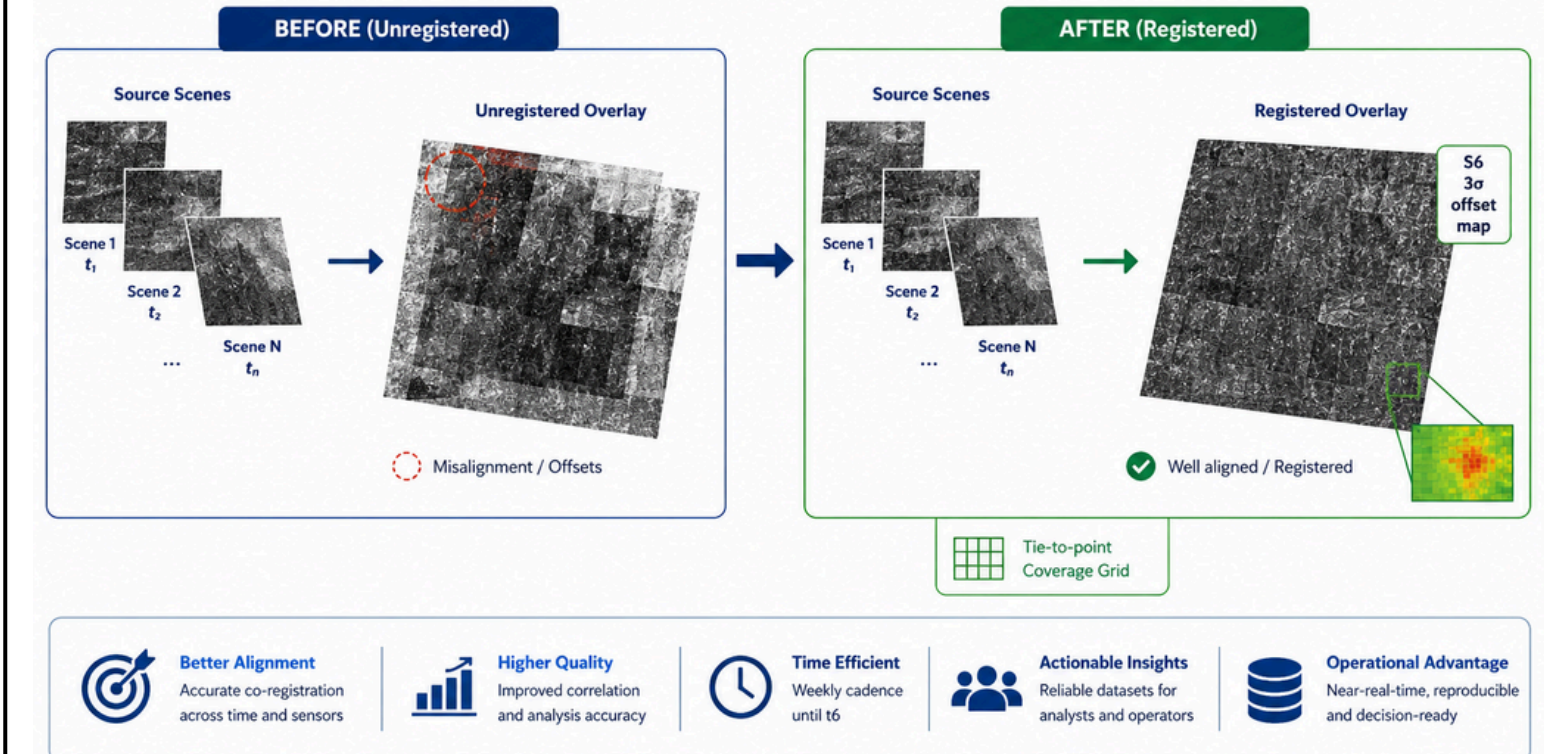
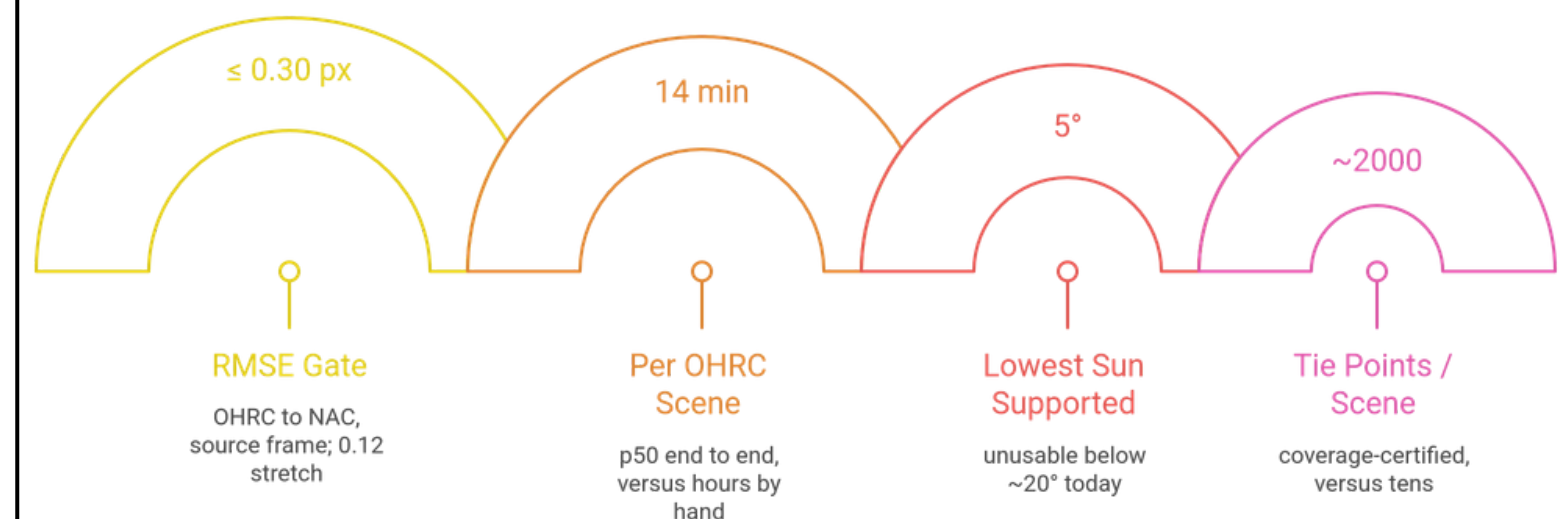
- Hours of manual tie-point picking become minutes unattended
- No licence cost: the entire stack is open-source
- Content-addressed caching means repeated renders cost nothing
- Ends duplicated manual effort across research groups
- Science return realised now, not after a re-processing

SUSTAINABILITY & LONGEVITY

- Air-gapped on-prem: no continuous cloud footprint
- Caching avoids recomputing the GPU-heavy render stage
- Version-pinned reference data keeps past work valid
- Dependency-free core outlives UI and orchestration
- Retention (3 y products, 7 y audit) is configuration, not code

ALIGNMENT & STANDARDS

- Serves ISRO / ISSDC archive use; LRO as lunar datum
- PDS4, ISIS, USGS CSM, SPICE, OGC COG, STAC 1.0, OpenAPI
- Feeds landing-site characterisation and change detection

CO-REGISTRATION PIPELINE**Overlay****Key Metrics for OHRC Scene Processing**

MISSION DATA & ARCHIVES

- Chandrayaan-2 OHRC 0.25 m, TMC-2 5 m, IIRS ~80 m / 250 bands — ISSDC PRADAN
- LRO NAC 0.5–2 m panchromatic — PDS Imaging Node / ASU
- PDS4 labels and reconstructed SPICE kernels — NAIF
- Chandrayaan-2 lunar images: <https://chmapbrowse.issdc.gov.in/>
- LRO NAC Images (Downloads): <https://lroc.im-ldi.com/images/downloads/>
- LRO QuickMap: <https://quickmap.lroc.im-ldi.com/>

REFERENCE TOPOGRAPHY & ALBEDO

- SLDEM2015 — merged LOLA and SELENE Terrain Camera global DEM
- LOLA gridded elevation; LRO NAC-derived DTMs where available
- LROC WAC global albedo and photometric parameter maps
- [add dataset versions and access links]

HOW THE RESEARCH SHAPED THE DESIGN

- Shadows carry no terrain information, so no descriptor can recover it → render instead of describe
- Scale is known from metadata → equivariance with a known factor, not invariance
- No public Chandrayaan-2 to NAC training corpus → generate it with the renderer
- Pixel locking is a systematic, not a floor → model and subtract the bias
- Uniformity is impossible in mutual shadow → define it over eligible cells only

PHOTOMETRY & PHOTOCLINOMETRY

- Hapke 2012 bidirectional reflectance model (w, b, c, B0, h, theta-bar)
- Opposition effect and phase-angle behaviour of lunar regolith
- Phase congruency as an illumination-invariant feature measure
- Shape-from-shading and slope-from-radiance inversion literature

PRIOR ART BENCHMARKED

- MoonMetaSync and IntFeat — SIFT / ORB hybrid lunar matching
- UR-SIFT, OS-SIFT, RIFT, Cof-SIFT — multi-modal descriptors
- OSDM-MReg-class learned multi-modal registration
- ISIS and Ames Stereo Pipeline control-network workflows

MATCHING & ESTIMATION METHODS

- LoFTR and RoMa detector-free transformer matchers
- MAGSAC++ robust estimation; ECC and Fourier sub-pixel estimators
- Pixel-locking bias in correlation-based sub-pixel refinement
- Submodular maximisation with greedy near-optimality guarantees

TOOLING & STANDARDS

- USGS ISIS 8, Ames Stereo Pipeline, usgscsm / CSM, NAIF SPICE
- Ceres Solver 2.2; GDAL 3.8; PyTorch 2.4
- PDS4, OGC Cloud-Optimized GeoTIFF, STAC 1.0, OpenAPI 3.1
- Clark-Evans index and Moran’s I for spatial diagnostics

Metric	SELENE-XR (Proposed)	SIFT-class	Learned Registration
Low-sun performance	Excellent / Robust	Poor (shadow sensitive)	Good (model-dependent)
Max. GSD ratio	Large (cross-sensor)	Small (<3x)	Medium (scale-aware)
Uncertainty output	Per-point covariance	Mask only	Heuristic confidence
Coverage guarantee	Uniform dispersion	Feature clusters	Architecture-specific
Relief handling	Rigorous DTM integration	Weak / Affine model	3D-aware